The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 December 1997

INCH POUND

MIL-PRF-19500/228H
14 September 1997
SUPERSEDING
MIL-S-19500/228G
21 May 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER,
TYPES 1N3611, 1N3612, 1N3613, 1N3614, 1N3957
JAN AND JANTX

These devices are no longer active for new design, preferred devices are 1N5614, 1N5616, 1N5618, 1N5620, 1N5622 on MIL-PRF-19500/427

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 1.0 ampere silicon rectifier diodes. Two levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 similar to DO - 41.

1.3 Maximum ratings.

Types	V_{RWM}	I_O	
		At: T_A = +100°C 1/ 2/	At: T_A = +150°C 1/ 2/
	V(pk)	A dc	mA dc
1N3611	200	1	300
1N3612	400	1	300
1N3613	600	1	300
1N3614	800	1	300
1N3957	1,000	1	300

1/ From I_O rating is independent of heat sinking, special mounting, or leads of the device.

2/ Derate linearly at 13.3 mA between T_A = +100°C and T_A = +175°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Maximum ratings. Continued

Types	I_{FSM} T_A = +25°C I_O = 1 A dc t_p = 8.0 ms	Barometric pressure (reduced)	T_{STG} and T_J	$R_{\theta JL}$ L = .375 inch
	A(pk)	mmHg	°C	°C/W
1N3611	30	8	-65 to +175	38
1N3612	30	8	-65 to +175	38
1N3613	30	8	-65 to +175	38
1N3614	30	54	-65 to +175	38
1N3957	30	87	-65 to +175	38

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

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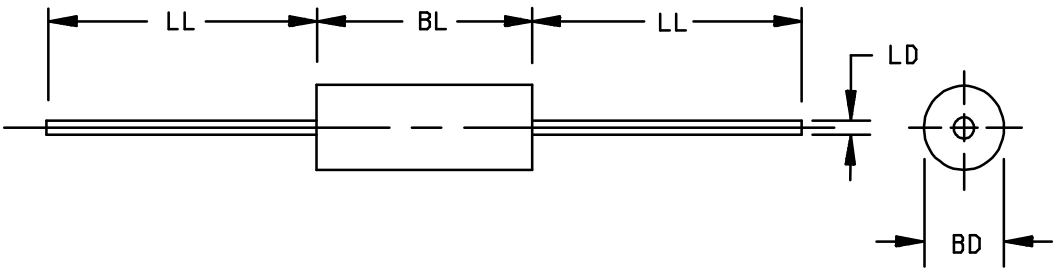

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.110	1.52	2.79	
BL	.140	.205	3.56	5.21	
LD	.025	.034	0.64	0.86	3
LL	.600	1.500	15.24	38.10	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension LD shall be measured at the largest diameter.
4. Dimension BD shall include the sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.

FIGURE 1. Physical dimensions.

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3.2 Associated specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein. No lead (Pb) shall be used in the construction of the die bonds.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein.

3.4.2 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature bonding between both sides of the silicon die and terminal pins.

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

3.6 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end or the use of other techniques considered commercial practice.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4. and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX level only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement JANTX level
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.2
13	Subgroup 2 of table I herein. ΔI_{R1} = 100 percent of initial reading or ± 50 nA dc, whichever is greater. ΔV_{F1} = ± 0.1 V dc.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B of MIL-STD-750, T_A = room ambient as defined in the general requirements in 4.5 of MIL-PRF-19500; V_{RWM} = full rated V_{RWM} (see 1.3); f = 50 to 60 Hz; I_O = 1 A; 96 hours.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, appendix E, table V, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, appendix E, table VIb of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	4066	I_{FSM} = 30 A(pk), 10 surges of 8.3 ms each at 1 minute intervals, I_O = 1 A dc, V_{RWM} = rated V_{RWM} (see 1.3), T_A = +25°C. Sampling plan = 10
B3	1026	T_A = +150°C ± 5°C, I_O = 300 mA, V_R = rated V_{RWM} (see 1.3), t = 340 hours, f = 60 Hz.
B5	3101 or 4081	+25°C ≤ T_L ≤ +35°C (recorded before test is performed), $R_{\theta JL}$ ≤ maximum rated $R_{\theta JL}$ (see 1.3).
B6	1032	T_A = +175°C, t = 340 hours.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition A, weight = 4 pounds, t = 15 s.
C2	2036	Test condition E.
C5	1001	V_{RWM} = rated V_{RWM} (see 1.3), pressure = rated pressure (see 1.3), t = 1 minute, Sampling plan = 15.
C6	1026	T_A = 150°C ± 5°C, I_O = 300 mA, V_{RWM} = rated V_{RWM} (see 1.3), f = 60 Hz.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Burn-in and steady-state operation life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180°, nor less than 150°.

4.5.2 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 1 A dc (pulsed) (see 4.5.2); duty cycle = 2 percent maximum	V_{F1}	0.6	1.1	V pk
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3) DC method	I_{R1}		1.0	µA dc
Subgroup 3						
High temperature operation:		T_A = 150°C				
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3) DC method	I_{R2}		300	µa dc
Low temperature operation:		T_A = -65°C				
Forward voltage	4011	I_F = 1.0 A dc (pulsed) (see 4.5.2)	V_{F2}	0.6	1.5	V dc

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 1.0 A dc pulse width = 8.3 ms maximum, duty cycle = 2 percent maximum	V_{F1}	0.6	1.1	V dc
2.	Reverse current	4016	DC method; V_R = rated (see 1.3)	I_{R1}		1.0	µA dc
3.	Forward voltage	4011	I_F = 1.0 A dc pulse width = 8.3 ms maximum; duty cycle = 2 percent maximum	ΔV_{F2}	± 0.1 V dc maximum		
4.	Reverse current	4016	DC method; V_R = rated (see 1.3)	ΔI_{R1}	± 50 nA dc or 100 percent of initial value, whichever is greater.		

1/ The electrical measurements for appendix E, table VIb (JANTX) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1, 2, 3, and 4.

c. Subgroup 6, see table II herein, steps 1 and 2.

d. Subgroup 7, see table II herein, steps 1 and 2.

2/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1 and 2.

c. Subgroup 5, see table II herein, step 2.

d. Subgroup 6, see table II herein, steps 1 and 2.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI, SM)
Navy - AS, CG, MC
Air Force - 13, 19, 70, 80, 85, 99

Preparing activity:
DLA - CC

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/228H	2. DOCUMENT DATE (YYMMDD) 970914

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER, TYPES 1N3611, 1N3612, 1N3613, 1N3614, 1N3957, JAN AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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